

AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation report

Single living document. Superseded claims are struck through, not deleted.

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Abstract

Mission. A working end-to-end AWI-ESM3 (FESOM2-CORE3-cavity + OIFS-48r1 + LPJ-GUESS + WAM) ↔ Antarctic-PISM cycle: awiesm3 year → PISM decade → awiesm3 year on a changed mesh, with all three coastline-following members (FESOM cavity, OIFS land-sea mask, WAM sea points) regenerated per leg.

Where we are. Five real bugs have been found and fixed (§1); the wave-model coastline is solved structurally (§2); the workflow performs genuine awiesm3→PISM→awiesm3 handovers (§1.5); and the recurring OIFS crash family was traced past three partial fixes to its true root: **the ice-atmosphere surface-temperature coupling architecture itself is unstable in the marginal ice zone** (§1.4). The atmosphere’s ice-tile skin was hard-pinned to the remapped FESOM ice temperature (skin conductivity 10^{10}), so kW-scale spurious fluxes circulated between the two models. The fix is architectural: OIFS now solves its own ice-tile skin implicitly each timestep, with slab conduction through the *coupled* FESOM ice thickness (new exchange field) and snow — the “coupled slab” (§3) — validated: the first slab run completed its year crash-free with the kilowatt flux events collapsed to zero.

Remaining open item: the too-weak Southern-Hemisphere summer sea-ice melt-back relative to the 500-year v3.4 reference (§4).

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1 The five bugs

1.1 Bug I — PISM was fed surface water as ice-base forcing

`coupling_ocean2pism` built PISM's `-ocean` `th` forcing as a mean over FESOM level *indices* 1–20 = **0–410 m including the summer surface** (the correct `-selevel,150/600` sat in dead code). PISM received ice-base water up to $+6.5^{\circ}\text{C}$ even from a healthy ocean, melted at $\sim 30\text{ m/yr}$, and shed **24% of its floating area per decade** — every mesh-change leg then had to absorb a catastrophic geometry increment. FESOM's own cavity melt in the same runs was sane (2.2 m/yr mean, 0.72 median).

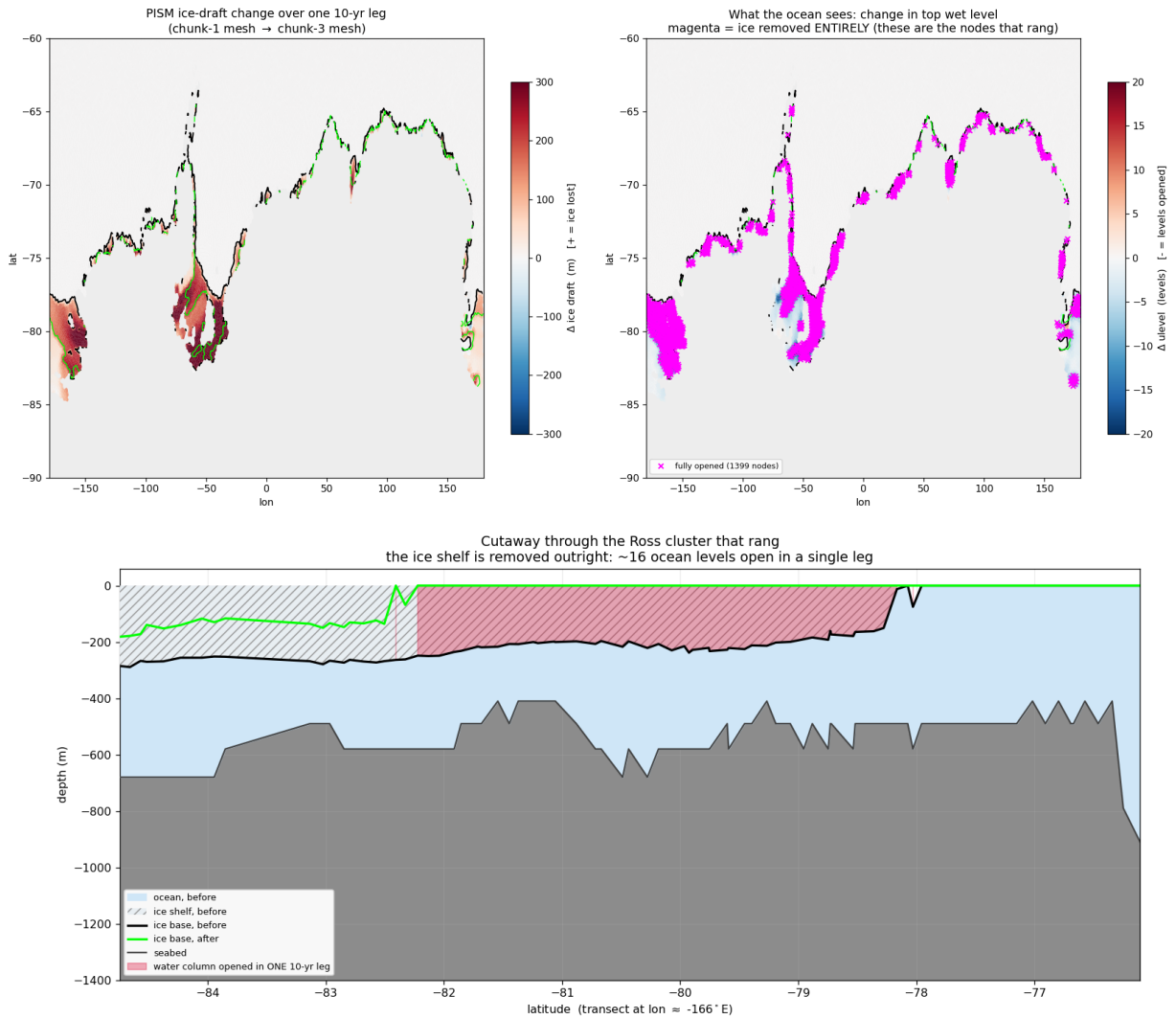


Figure 1: What Bug I did to PISM in one 10-year leg under surface-water forcing: 1399 nodes lose their ice entirely (magenta ring around the whole margin), the Ross front retreats $\approx 440\text{ km}$ (section: ice base $-200 \dots -280\text{ m} \rightarrow$ open ocean across four degrees of latitude).

Fix (93a207ee): the DIRECT route. PISM consumes FESOM's own cavity fluxes via `-ocean` given: `fw` $\rightarrow$ `shelfbmassflux` ($\times 1000$, sign preserved), `sst` $\rightarrow$ `shelfbtemp`. One model owns the ice-ocean interface. The `-ocean` given machinery existed all along but silently fell back to `th` because

its input is a FESOM1.x-era bundle FESOM2 never wrote; `fesom2ice` now synthesizes it (plus three exhumations of never-run 2D-path code: 3f07d7a2, 74671c27, bb9d00c2).

1.2 Bug II — the exchange weights scrambled the coupled ocean

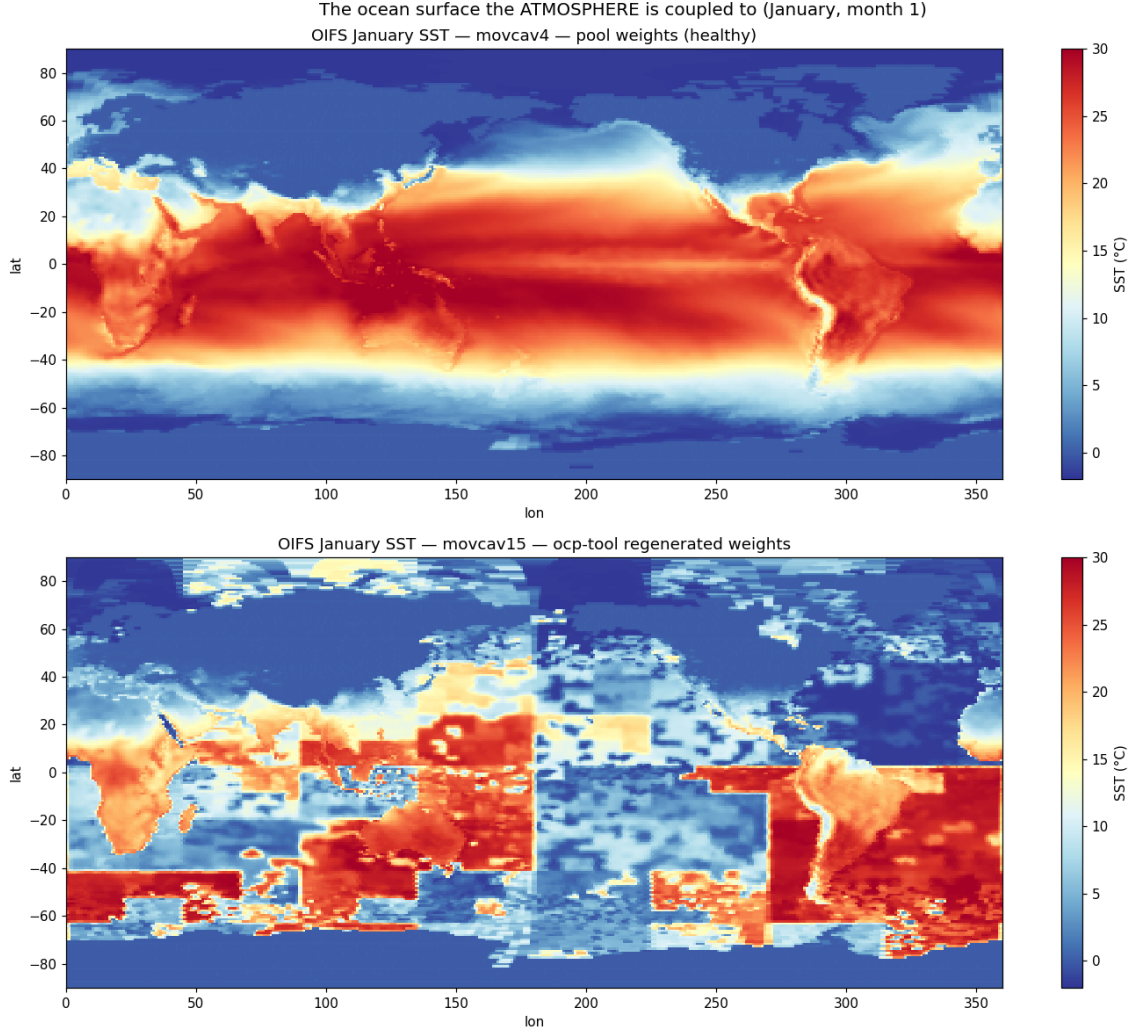


Figure 2: **The picture that settles it.** January SST as OIFS holds it. Top: pool weights — textbook. Bottom: offline-engine weights — a *tile-translocated jigsaw*: tropical blocks at 55°S, polar blocks on the equator, a seam at lon 180 where the node ordering breaks.

FESOM exchanges coupling fields in *partition* (`my_list/rank`) order; the offline weight engine (2026-06-17) built the feom geometry in *nod2d.out* order (median displacement 58°). Every submesh exchange was geographically tile-translocated (Fig. 2): OIFS saw Weddell ice fraction 0.04 while FESOM had 0.95; global sea-ice death followed within months.

Fix (c21fe7c2): `permute_feom_to_runtime.py` reorders the OASIS geometry to runtime order (via `dist_<nproc>/rpart.out`); `validate_feom_weights.py` applies every weight file to real latitudes and **aborts the leg** above 1° median error (nod2d-ordered weights fail at 42–58°, correct ones pass at $\leq 0.025^\circ$).

1.3 Bug III — the OASIS coupling restart was compound-scrambled

`remap_oasis_restart.py` gathered in nod2d space, but its input and required output are both *run-time*-ordered: the emitted `rstos.nc` matched *neither* ordering (correlation of `sst_feom` with the

restart’s own surface temperature: +0.15). OIFS’s first coupling interval saw 305 K water under polar air — a one-hour global jigsaw shock that nondeterministically detonated the convection scheme (identical inputs crashed at hour 19, hour 458, or never).

Fix (abe56ee2): un-permute via the old mesh’s `rpart.out` → gather → re-permute via the new mesh’s; correlation now +0.995. Pool `rstos.nc` rebuilt.

1.4 Bug IV — the ice-surface-temperature coupling is architecturally unstable

This is the mid-run OIFS crash family (`cubasen/vsurf/voskin` floating overflow, random timing, polar/MIZ ranks). It survived three partial fixes before the root was found; the full chain is worth recording because each layer was real, just not sufficient.

Layer 1: MIZ aliasing of the sent temperature. FESOM sent raw `ist`; ice-free nodes send the seawater freezing point, so the GAUSWGT cell blend in the marginal ice zone was dominated by warm open water: OIFS’s ice tile saw a surface tens of K too warm and returned a strongly negative flux to the fully-iced nodes. Partial fix: concentration-weighted `ist` (`ist·a_ice ÷ ci` on ingest, the EC-Earth/NEMO convention behind `ECE_CPL_NEMO_WEIGHTED_ICE`). Runs survived $3\times$ longer, still crashed.

Layer 2: the FESIM solve has no flux feedback within the coupling interval. FESIM’s skin solve relaxes toward $t^* = T_f + a2ihf \cdot zsniced/con$ with the received flux *frozen* for the hour and no dQ/dT term: under insulation the equilibrium runs away (measured floors 145–183 K; excursions to −954 K in one run). Partial fix: an implicit flux linearization anchored at the transmitted temperature, $Q(t) = a2ihf + \lambda(t_{ref} - t)$ with $\lambda = 4\varepsilon\sigma t_{ref}^3 + \lambda_{turb} \approx 20 \text{ W/m}^2/\text{K}$ (the NEMO-family fallback derivative), energy-closed by booking the linearization heat to the ice growth budget (fesom branch `awiesm3-implicit-ice-surftemp`, 38a558b5). Dips became bounded and self-recovering in most runs — but not all.

Layer 3 (the root): the atmosphere’s ice skin was never solved at all. Send-side instrumentation of `A_Q_ice` (term decomposition + tile state at every $|Q| > 1 \text{ kW/m}^2$ event) delivered the verdict:

- The received flux at cold FESOM nodes is **unphysical and anti-braking**: mean -15.8 kW/m^2 below 150 K (100% negative), -4.5 kW at 150–180 K, recurring all year with the MIZ seasonal cycle — at **49°N on 0.1–0.5 m ice with thin snow** (Okhotsk / St. Lawrence). Insulation is irrelevant there; no realistic λ brakes kilowatts.
- At the OIFS send, the generating cells have ice-tile fractions down to 0.04 and **tile skin temperatures of ~204 K under ~275 K maritime air**; the sensible + latent terms then reach $+17.8 \text{ kW/m}^2$ per tile area. With weighted-`ist` the tile skins were *still* 189–204 K at substantial tiles — the cold temperature is genuine FESIM state, faithfully communicated.
- The reason the skin can sit 80 K below the air: `vlamsk_mod.F90`, sea-ice tile, with `LNEMOSICOU= .true.` (the default): `PLAMSK = 1e10` — **the skin conductivity is set to infinity, pinning the tile skin to the prescribed (remapped FESOM) ice temperature**. The atmosphere’s surface scheme was forbidden from solving its ice surface; it computed fluxes against whatever arrived through the remap.

The loop, in one line: FESIM’s oscillating solve → remap → *pinned* OIFS skin → kW fluxes against an unresolvable surface → remap → FESIM slams further — a two-model positive feedback closed through the marginal ice zone, metastable enough that identical configurations coin-flipped

between completing a year and dying at day 2 (runs movcav25–31: raw/weighted $\times \lambda = 3.4/20 \times$ dirty/clean initial state — every combination crash-prone).

Root fix: the coupled slab (§3). Stop prescribing the atmosphere’s ice skin altogether; let OIFS solve it implicitly each timestep with conduction through the *coupled* FESOM ice and snow thickness. This removes the loop by construction while keeping the flux honest about the actual ice state. Upstream note: mainline AWI-CM3 carries the same pinned-skin architecture and therefore the same latent (if rarer) crash mode.

1.5 Bug V — iterative coupling never handed over to PISM

The awiesm3→PISM model switch is resolved at SimulationSetup init by reading the `chunk_date` file, whose path `esm_runscripts/chunky_parts.py` builds by *raw string concatenation of `general.base_dir` before `esm_parser` resolves variables*. A `base_dir` containing `${user}` was probed verbatim; the silent `isfile()` miss fell back to “chunk 1, model 1” — awiesm3 self-resubmitted to its own final date and PISM never ran. Proven by log comparison: identical resubmission command lines resolve to `Valid Model Names = ['pism']` under a literal `base_dir` (movcav16 era) and to `['fesom', 'oifs', ...]` under `${user}`.

Fix (d61297e8): `_early_resolve()` expands `${...}` against `general` (environment fallback) in all three chunk-file path builders. Regression-tested against a real chunk file, and validated live: the first genuine awiesm3→PISM handover of the campaign (movcav27), and a full awiesm3→PISM→couple_out→chunk-3-couple_in chain (movcav29).

2 The wave model’s frozen coastline (stage-6 blocker, solved)

Chunk-3 died deterministically at its first physics step in `voskin_mod.F90:282` squaring WAM’s Stokes drift. WAM has no runtime tiling: its sea-point set is baked into `wam_grid_tables` at preprocessing time and had never been regenerated. A cell that is ocean per the regenerated lsm but land per WAM never receives wave data; the Stokes/Charnock fields are srf-restart-carried, so a *warm* start after a coastline change reads never-written storage (cold starts zero-initialize — why chunk-1 tolerates its 12 mismatch cells and historical lsm changes never hit this). Chunk-3 had 125 mismatch cells. WAM itself was separately exonerated for the Bug-IV crash family (a WAM-off run crashed identically).

Fix: one-time maximal-ocean tables (option A; tolerance patching rejected). WAM’s sea-point set was rebuilt from ETOPO1 with the southern limit extended $78 \rightarrow 85^\circ\text{S}$, then *max-ocean edited*: every cell the FESOM max-mesh can make wet — open ocean or sub-shelf cavity — is forced to sea at its FESOM bottom depth (only ever adding ocean). 1277 cells opened ($29\,555 \rightarrow 30\,832$ sea points), 283 of them south of 55°S : exactly the band a moving coastline walks through (Fig. 3). The tables never change again, wave restarts stay index-compatible across all future mesh changes, and WAM cold-starts once at the switch leg. Known cosmetic debt: the max-ocean edit did not update the obstruction coefficients.

3 The coupled-slab architecture

The new division of labour: **OIFS owns the ice surface temperature; FESOM owns the ice state**. OIFS’s surface scheme solves the ice-tile skin implicitly within every atmosphere timestep (full radiative + turbulent dQ/dT , finite skin conductivity), with slab conduction through the coupled per-ice thickness and snow depth. FESOM’s albedo still drives the radiation; FESOM’s own `ist` solve remains for its internal state (albedo, melt ponds) but is read by nobody — the feedback loop is cut by construction. The machinery is EC-Earth/KNMI heritage (ICESTATENEMO, 2008; the SI3 branch of `callpar`) — our contribution is wiring the FESOM branch, which lacked only the thickness.

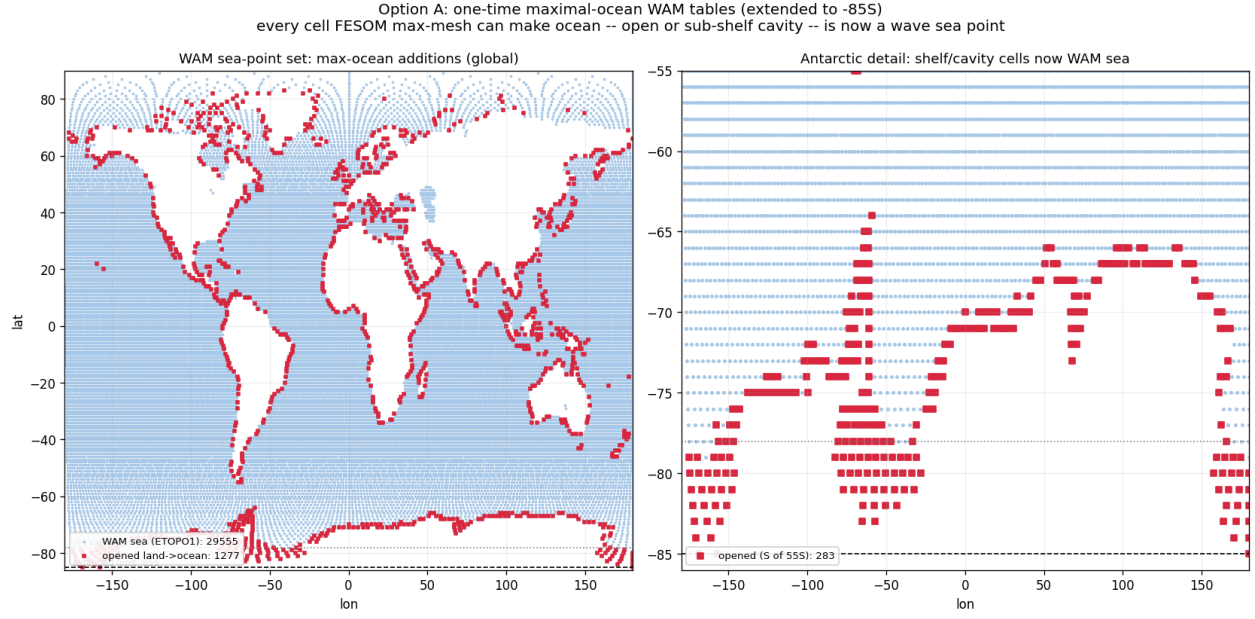


Figure 3: One-time maximal-ocean WAM tables (extended to 85°S). Blue: wave sea points from ETOPO1. Red: cells opened land→ocean because the FESOM max-mesh has a water column there. Pre-registering them makes the `voskin` step-0 crash structurally impossible on any future leg.

New exchange field. FESOM sends `sit_feom` = effective (grid-mean) ice thickness `m_ice` (o2a collection, `A_Ice_thickness`); `ECE_FESIM_GET_ICE_STATE` divides by the received ice fraction (standard ε -guard) for the per-ice thickness the slab needs, falling back to the climatological 1.5 m only where the cell has no significant ice. Snow thickness (`snt_feom`) was already exchanged and now actually reaches `surfseb` via `SURFL%ZSNTICE`.

The switch set (verified in code, not inferred):

switch	setting	verified effect
LNEMOSICOU	.false.	<code>vlamsk</code> : sea-ice tile skin conductivity becomes <i>finite</i> (land-tile-style lookup) instead of 10^{10} — the skin is genuinely solved by the SEB. The prior default pinned it to the prescribed ice temperature; this pin was the kilowatt engine of Bug IV.
LNEMOLIMTEMP	.false.	disables the hourly overwrite of OIFS’s 4-layer ice temperature (<code>SP_SB YTL</code>) from the FESOM ist ingest; the profile is OIFS-prognostic (ICMGG-initialized, restart-carried). FESOM’s ist send becomes inert.
LNEMOLIMTHK	.true.	activates the <code>callpar</code> fill of <code>ZTHKICE/ZSNTICE</code> from the coupled fields — the skin/top-layer conduction feels the real ice thickness and snow insulation. (The deep 4-layer discretization keeps its fixed depths; the skin conduction is the term that controls winter fluxes.)
LMCCDYNSEAICE	.true.	master enable; false force-resets the whole <code>LNEMOLIM*</code> family (<code>sumcc</code>).
LNEMOLIMALB	.true. (default)	FESOM’s albedo drives OIFS radiation (<code>surfrad_layer</code> → FESIM getter; raw–raw convention, consistent).
ECE_CPL_NEMO_WEIGHTED_ICE	.true. (inert)	only acted inside the disabled ist ingest.
FESOM <code>latm_owns_ist</code>	.true.	FESIM growth budget takes the atmosphere flux directly (<code>Qatmice=-a2ihf</code> , the ECHAM convention); <code>qres/qcon/qlam</code> booking off.

The `sumcc` guard “LNEMOSICOU without LNEMOLIMTEMP does not make sense” correctly enforces that a pinned skin requires a prescribed temperature; our mode flips both off. In one sentence: *before, OIFS’s ice surface was a mirror bolted to whatever FESOM’s oscillating solve sent; now it is a real solved surface conducting through FESOM’s actual ice and snow.*

Validated (movcav33). The first coupled-slab run completed its year crash-free and handed over to PISM. With the instrumentation still armed: kW-scale send events at real ice tiles went from 1410–1930 per run to **zero** (117 residual events are all phantom slivers with ice fraction $\sim 10^{-7}$, a `ZMASK` threshold cosmetic); event tile skins moved from 189–212 K into the physical 247–265 K range (Fig. 4); the fluxes FESOM receives at its coldest nodes are $-63 \dots -80 \text{ W/m}^2$ (textbook polar values, was -10 kW), and FESIM’s internal temperature tracks its anchor to $< 0.1 \text{ K}$.

4 State of the cycle and of the climate

4.1 What is proven

- **Chunk-1 is quantitatively healthy** (movcav16-era checklist, all pass): January SST physical, zero translocated cells; OIFS Weddell September ci 0.61/0.99 (was 0.04); both-hemisphere sea-ice annual cycles; cavity melt 2.16 m/yr mean (reference 2.2; jigsaw 28) and thermal driving $+0.18 \text{ K}$ (reference ~ 0.2 ; jigsaw $+1.25$).
- **The live mesh change works:** 208 810 → 211 174 nodes, weights regenerated and runtime-order-validated at $\leq 0.024^\circ$, restarts remapped through the `rpart` sandwich.

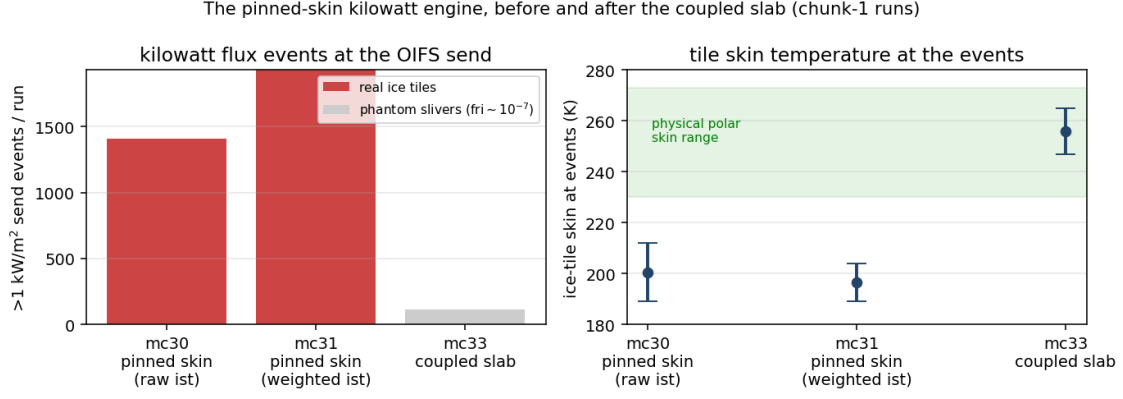


Figure 4: The kilowatt engine, before and after. Left: $>1\text{ kW/m}^2$ events at the OIFS send per chunk-1 run. Right: the ice-tile skin temperature at those events — pinned-skin runs sit 60–80 K below the physical range; the solved skin cannot.

- **The PISM decade runs on FESOM’s own melt** (grounded -0.10% , floating fringe -21 of 133 cells) and **the workflow hands over in both directions** (Bug V fixed; movcav29 ran awiesm3 $\rightarrow$ PISM $\rightarrow$ couple_out $\rightarrow$ chunk-3 couple_in unattended).

4.2 Open item: the Southern-Hemisphere summer melt-back

The ice-thermodynamics code is byte-identical to the v3.4 reference; the difference is state and forcing. Ours keeps $\sim 2.5\times$ the reference’s winter SH extent and only a $\sim 6\times$ summer retreat where the reference manages $\sim 200\times$ (Fig. 5), and piles 2–3.5 m of snow against the Antarctic ice-shelf coast (Fig. 6). Two mitigations exist: a *clean chunk-1 ice initial state* (reference year-1849 fields indexed onto the submesh via `map_nod.out`, 7 Ronne artifact nodes hybridized; snow max 0.81 m — in use since movcav29), and the coupled slab itself (the pinned-skin era’s fictitious equilibria drove a spurious congelation-growth term). Whether these restore the reference’s melt-back is the first climate question for the post-slab runs; meltponds were checked and are *not* the SH lever (the `hsno>hs1` lockout makes old and corrected schemes identical there).

4.3 The restart-seam arc (historical, resolved by cold-start + clean ini)

The original FESOM cavity-margin blowup: a remapped warm restart mixes an evolved, balanced dynamic state (98% of nodes, bit-exact) with patched values at changed nodes; the *seam* between them is the instability. The decisive control:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up restart	stable, worst CFLz 2.61
PISM cavity, cold	1200 s	cold start	stable, CFLz 3.44
PISM cavity, warm	1200 s	remapped restart	dies day 4.0 (CFLz 9.2)
PISM cavity, warm	60 s	remapped restart	dies day 4.4 (η_n , 0 CFLz)

The ocean lives happily in PISM’s cavity; the production answer became: run chunk-1 cold-started on the PISM cavity (artifacts pooled), and let `remap_restart` carry only genuine per-leg increments. The first real increment test crashed on 17 fully-opened columns (ice removed entirely, 16 levels at once) out of 1399 — but its numbers are **TAINTED** (measured under Bugs I+II) and the finding that matters survived into the fixes above: absorbing 99% of a geometry change is not a passing grade when the remaining 1% is fatal.

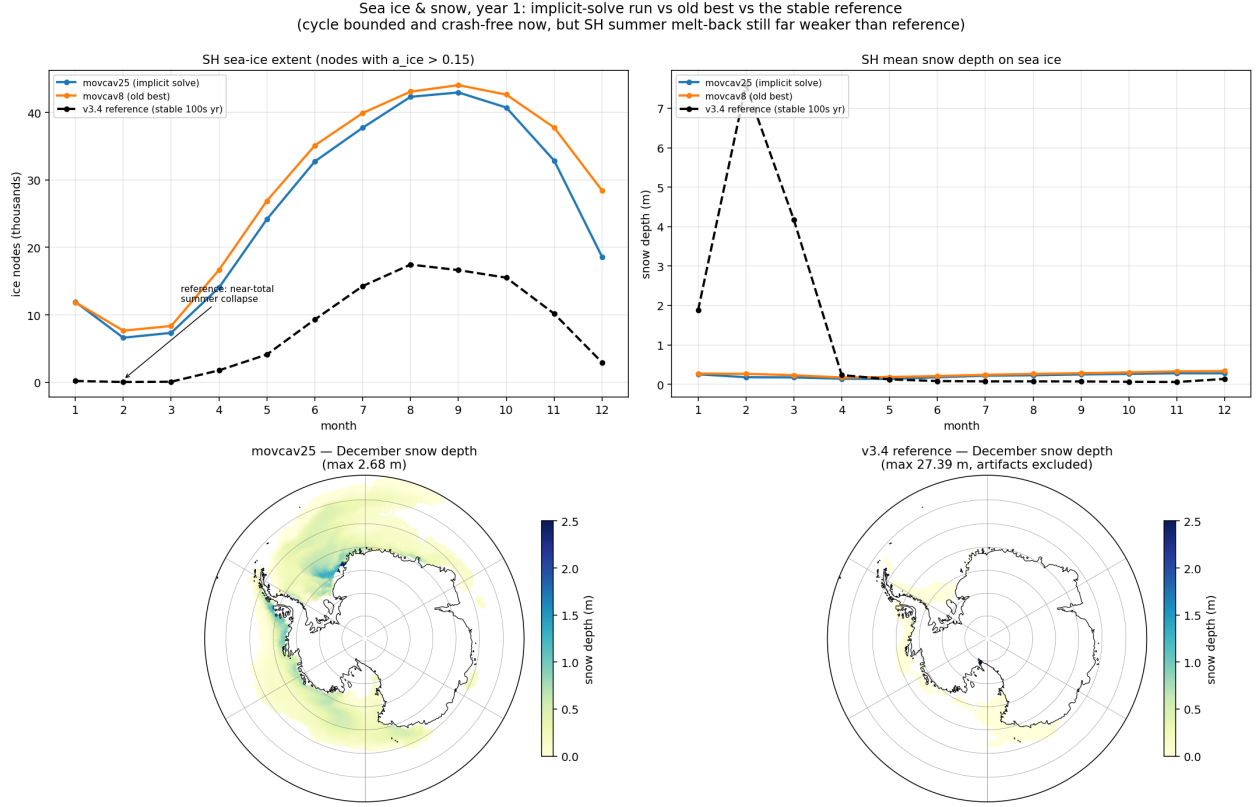


Figure 5: Year-1 sea ice and snow: implicit-solve run (blue) vs the movcav16-era state (orange) vs the 500-year-stable v3.4 reference (black dashed). Our runs track each other — the crash fixes did not degrade climate — but the reference collapses its SH ice almost completely each summer while ours retains $\sim 7k$ nodes, and its winter pack carries 0.03–0.08 m of snow against our 0.2–0.3 m. Bottom maps (same scale): December snow depth — our thick band hugs the Antarctic coast; the reference is nearly bare.

5 Secondary bugs, fixed along the way

Bug	Fix	Commit
Remap extrapolated unboundedly below the old seabed (manufactured -45°C) when columns deepen	take water from the nearest old node wet at that depth	eef4f6db
Sub-ice $\eta/hbar$ kept open-ocean values at newly-sub-ice nodes	zeroed (real bug, not the cure)	b17fb5c8
First mesh-change leg read the mother mesh as cavity-free (<code>cavity_nlvls.out</code> missing from the <code>mother_as_old</code> links) $\Rightarrow$ zeros in newly-wet levels, mstep-1 blowup	link the cavity files	functions
Same-chunk <code>couple_in</code> rerun promoted its own submesh to <code>previous_submesh</code> (old = new $\Rightarrow$ zeros again)	rerun guard	b9c1e370
OASIS GSSPOS conservation blowup on near-zero target residuals	<code>gauswgt_gsmart</code> transform	3e94d4da

Bug	Fix	Commit
namcouple feom dimension never actually patched (env var from another subjob's shell)	fixed variable passing	3cf29e9e
fesom2ice loaded the static full mesh against submesh output	load the run mesh	63cfcc91
previous_submesh read but never created	created; plus restart size guard	3b44a704
Failed coupling functions did not stop the workflow (backgrounded, observe saw no exit code)	.coupling_failed sentinel honoured by observe	7953585b
rstos size guard read the wrong dimension	fixed	61c8ed8a
ocp-tool flipped a coastline cell's mask+soil only, leaving an inconsistent surface column	masked nearest-neighbour rebuild of the full column	ocp-tool PR #51

The recurring pattern: the guards were right and the plumbing around them defeated them — a correct refusal followed by an unconditional continue is worse than no guard at all.

6 Falsified claims (kept, struck through)

- ~~Geostrophically unbalanced remapped state; initialize changed cells in thermal wind balance.~~ Failed its own offline gate (discrete ∇p noisier than NN fill); ships opt-in only, off by default.
- ~~The sub-ice ssh BC violation is the blowup.~~ Real bug, fixed, run still died.
- ~~The per-leg PISM increment is gentle (≈ 22 m mean).~~ The mean was diluted by the 98% unchanged domain; the distribution is wildly skewed (max 923 m, 1399 fully-opened nodes).
- ~~The salt plume parameterization being off explains the warm cavity.~~ The spp-off cavity is the *coldest* in the CORE3 reference runs; the warm cavity was Bug II.
- ~~The native ICMGG causes the mid-run crash family.~~ The step-0 swap tests proved the native, coastline-following surface set is *required* on the submesh (62 difference-zone cells cannot initialize otherwise); the mid-run family was Bug IV.
- ~~Weighted-ist cures the crash family.~~ Necessary hygiene for the o2a blend, not sufficient: the pinned OIFS skin regenerated the instability from bounded inputs.
- ~~The insulation/snow pile-up is the crash root.~~ It raises the loop gain (and is a real climate bias, §4), but the kW events occur on *thin* ice at 49°N; the clean-ini run crashed identically.

7 Method notes

- Cheap controls first.** The cold start (geometry vs state), running the fix, plotting the distribution instead of the mean — each killed a confident claim in minutes.
- Measurements from a corrupted run are worse than none** (Bug II poisoned Bug I's diagnosis for a week).
- “Same code as the stable reference” localizes nothing by itself** — the reference ran the same FESIM solve for 500 years; the difference was the state and, one layer deeper, a switch default that pinned a skin nobody remembered was pinned.

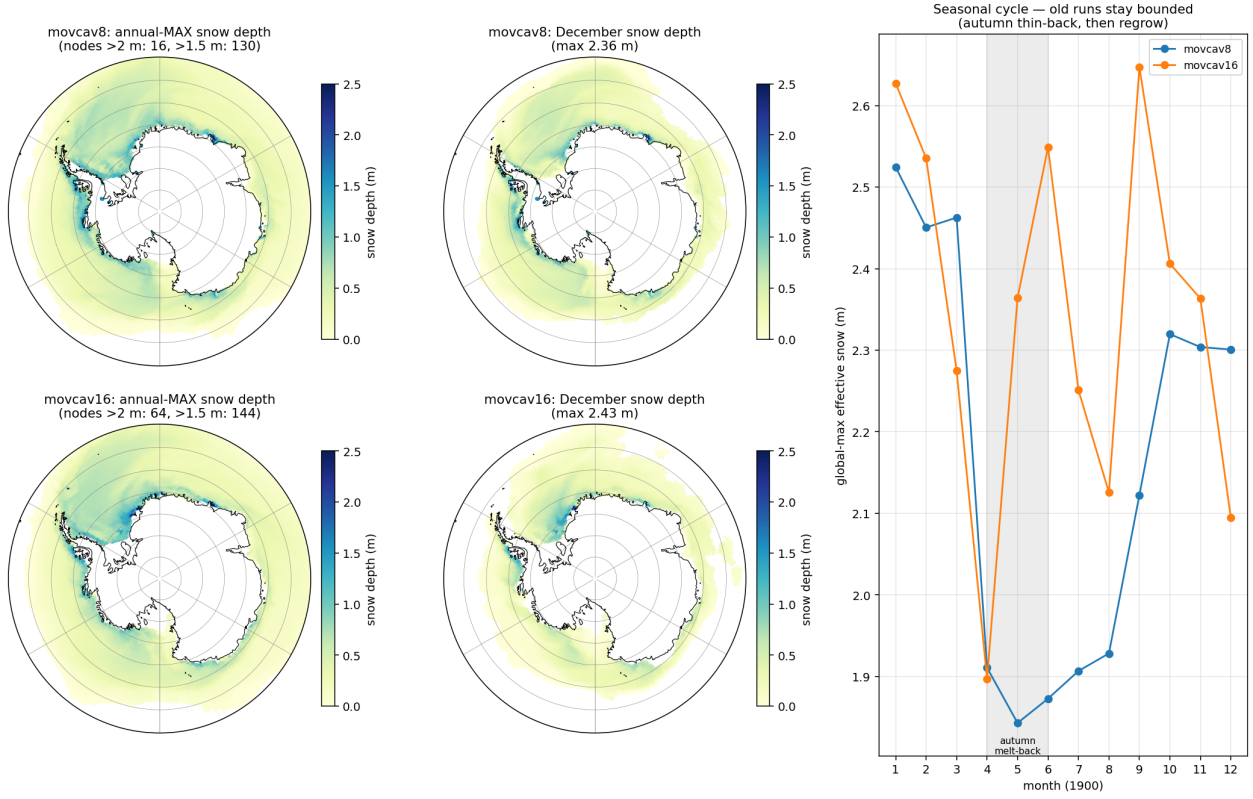


Figure 6: The extent of the coastal pile-up in the (pre-fix) movcav runs: annual-max and December snow depth, and the seasonal cycle. The thick snow is a band hugging the ice-shelf front (eastern Weddell/Fimbul, Amundsen), bounded at 2–2.6m by an autumn melt-back. Later runs that lost this melt-back drifted monotonically to 3.5 m — the insulation that raised Bug IV’s loop gain.

4. **Instrument both ends of a coupling before theorizing about either.** Receive-side logging (ICEDBG) proved the flux was corrupt; send-side decomposition (SENDDBG) named the term and the tile state in its first twelve lines. The two probes cost one rebuild each.
5. **Beware silent fallbacks** (Bug V’s `isfile()` miss; the `-ocean` given silent fallback to `th`). A fallback on a nonexistent path should say so.

8 Assets

- Send/receive flux instrumentation: SENDDBG (OIFS `A_Q_ice` decomposition + tile state), ICEDBG (FESIM received flux + state at cold nodes), CPLDBG (OIFS ingest bounds) — all log-only, strip before merging.
- `oasis_expout_o2a` runscript flag: every o2a exchange dumped to netcdf (~20× slowdown; crash-window probes only).
- Clean chunk-1 ice initial state (`prep/clean_ice_ini`): reference year-1849 ice fields indexed to the submesh; `map_nod.out` makes such remaps exact.
- A 2-minute offline reproducer (standalone FESOM on the PISM-cavity submesh) and a stable full-mesh control.
- `validate_feom_weights.py`: aborts any leg whose exchange weights are in the wrong node ordering — would have caught Bug II on day one.

- `verify_balance.py`: offline balance/stability gate for remapped restarts.
- Chunk-1 artifacts pooled at `input/fesom2/pism_cavity_ini/` (login-node launches).